GENOMIC SURVEILLANCE OF SURGICAL SITE INFECTIONS & AMR · UGHE–HMS

Lab 1

Files, Formats and Your First Galaxy History

Day 1 · 10:45–12:15 · 90 min
Pairs: driver + navigator · swap at Part C

LO2, LO3: recognise genomic file formats, read a quality score, and run a tool in Galaxy

Open your handout

Lab 1 handout

`Lab1_files_formats_galaxy.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

Plan for this lab

Part	What	Time
A	Files and folders	15 min
B	Looking inside the formats	25 min
C	Your first Galaxy history	40 min
	Wrap-up	10 min

Before you start

- Log in to `usegalaxy.eu`
- **Driver:** hands on the keyboard
- **Navigator:** reads the handout and spots mistakes

> Stuck or have a question? **Raise your hand** and the instructor will come to you.

What's in a file name?

`isolate01_R1.fastq.gz`

`isolate01`	`R1`	`.fastq`	`.gz`
sample name	read 1 of a pair	format	compressed

CHECK YOURSELF

What do you expect is inside `UGHE-SSI-0042_R2.fastq.gz`?

Two more things to know

- **Plain-text files** (FASTA, FASTQ, CSV) are just letters: any text editor can open them
- **Genomic files are big**: 200 MB–1 GB per isolate

KEY MESSAGE

That’s why we analyse in Galaxy, on computers on the internet, not on our laptops.

FASTA: just sequence

```
>contig_1 length=120
TTCGCTCTATTGACTACGACGCGCTCATTCCCTTGTCGGAGAGTTATGGAACAAGGACGC
TGTCTGAGACTAGAAGACAGATAGTGCACACGACCGGCGTCGGAGAAACTCTATTTGCCG
>contig_2 length=60
CCTGACAAGTCAATGCGATCCGTAGGGGCAGCGCAGTATGCCAAGACTATAGGCACTGTC
```

- Each entry starts with > and a **name**; the sequence follows
- **Assemblies** (genomes) are stored in FASTA

CHECK YOURSELF

How many sequences? How many bases in total?

FASTQ: sequence + quality (raw reads)

Every read takes **exactly 4 lines**:

Line	Example	Meaning
1	`@SSI_isolate_01_read1`	@ + read name
2	`GCTAAAGACAATTACATAACATACACGTCAGCACGAAACT`	the DNA sequence
3	`+`	separator
4	`GHHGFGHIIIIIFIGGHGIFIIGHIHIGHIIGFHHDACBC`	quality, one symbol per base

Reading a quality score

Symbol	#	+	5	?	I
Q score	2	10	20	30	40
Chance it's wrong	≈ 63%	1 in 10	1 in 100	1 in 1,000	1 in 10,000

RULE OF THUMB

Capital letters (A–J) and ? or @ are good (Q30 and above).
Symbols like #, $, %, & are bad (below Q10).

Check yourself: read 4

```
IIHFGIIIFIIIIIFIIHFIIIIHI#'&'##&%''''$(%
```

CHECK YOURSELF

1. Which part of read 4 (start or end) has poor quality?
2. What should we do with those bases?

CSV / TSV: tables

```
isolate_id,organism,collection_date,ward
UGHE-SSI-0001,Klebsiella pneumoniae,2026-03-02,Maternity
UGHE-SSI-0002,Escherichia coli,2026-03-05,Surgical
```

- One row per isolate, one column per piece of information
- Commas (CSV) or tabs (TSV) separate the columns
- They open in Excel or Google Sheets

Swap driver and navigator

The navigator takes the keyboard; the driver reads the handout

A tour of Galaxy

- **Left, Tools:** search box and tool list
- **Centre:** tool forms and results
- **Right, History:** your files (**datasets**), in order

Colour	Meaning
green	finished
yellow	running
grey	queued
red	error

Create a history and upload data

1. History panel → **“+”** → rename it `Lab1 – formats`
2. **Upload** → **Paste/Fetch data** → paste the **4 reads** from your handout
3. **Type:** `fastqsanger` · **Name:** `tiny_reads.fastq`
4. **Start** → **Close**
5. When it turns **green**: click the name, then the **eye icon**

PART C4

Run your first tool: count lines

1. Search the tools: **Line/Word/Character count**
2. “Text file” → `tiny_reads.fastq`
3. **Run tool**, then view the result (eye icon)

> **CHECK YOURSELF**
>
> How many lines? How many reads is that? *(4 lines per read)*

PART C5

Run a real QC tool: FastQC

1. Search the tools: **FastQC** → input `tiny_reads.fastq` → **Run tool**
2. Open the **Webpage** output (eye icon)
3. Find **“Per base sequence quality”**

CHECK YOURSELF

- Does quality go up or down towards the end of the reads?
- Would you trust a report based on only 4 reads?

PART C6

Your history is a lab notebook

- Click a dataset, then the **“i”** (Dataset details) icon
- Galaxy records the **exact** tool, version and settings

> **KEY MESSAGE**
>
> **This is reproducibility: anyone can see and repeat what you did.**

Wrap-up

Format	Holds
FASTA	sequences (assemblies)
FASTQ	reads + quality (raw data)
CSV / TSV	tables (metadata, results)

Upload → **Choose tool** → **Run** → **View**

After lunch: **real** reads from a bacterial isolate.

Adapted from the Galaxy Training Network “Introduction to Galaxy” and “Quality Control” tutorials (CC-BY 4.0).